7th Grade Math · Weekly Check · Unit 3 — C-NSO.2.3

Benchmark: **MA.7.NSO.2.3** · Real-world problems with rational numbers Student code: _____ Calculator: yes

Learning goal: I can solve real-world problems with rational numbers. (circle your level after scoring)

4	I can solve a multi-step real-world problem with rational numbers and explain whether the answer is reasonable.
3	I can solve a real-world problem involving rational numbers and state the answer with units.
2	I can pick the operation a word problem needs.
1	I need help to start this benchmark.

1. A tank contains $76\frac{3}{4}$ gallons of water. Water leaks out at 0.25 gallons per hour for 5 hours, and then $14\frac{1}{2}$ gallons are added.

How much water is in the tank now?

2. A group of 5 friends owes $83.60 for a canoe rental. They split the amount owed equally, and each friend's share is recorded as a negative balance. Then Jamal pays $\frac{3}{8}$ of his share.

What is Jamal's balance now?

Turn over →

3. At 6 a.m. the temperature at a weather station was -5.2°C. The temperature then fell 3.5°C each hour for 6 hours. Later in the day it rose 4.1°C.

What was the temperature after all of these changes?

4. The price of one share of a company's stock changed by these amounts over 4 days.

+$1.85, −$1.30, −$0.65, −$2.30

What was the mean (average) daily change in the price?

5. The temperature outside a cabin in January is 9.3°F. The temperature drops $3\frac{1}{2}^\circ\text{F}$ each hour.

After how many whole hours will the temperature first be below -3°F?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-NSO.2.3 · MA.7.NSO.2.3 · Real-world problems with rational numbers

Unit 3: Operations with Rational Numbers. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **90 gallons**

Water lost: $5 \times 0.25 = 1.25$ gallons.

After the loss: $76\frac{3}{4} - 1.25 = 75\frac{1}{2}$ gallons.

After adding water: $75\frac{1}{2} + 14\frac{1}{2} = 90$ gallons.

2. Regular **−$10.45**

Each share: $-\$83.60 \div 5 = -\$16.72$.

Paying $\frac{3}{8}$ of the share leaves $1 - \frac{3}{8} = \frac{5}{8}$ of it: $\frac{5}{8} \times (-\$16.72) = -\$10.45$.

Jamal's balance is −$10.45.

3. Regular -22.1°C

Total fall: $3.5 \times 6 = 21^\circ\text{C}$.

$-5.2 - 21 = -26.2$.

$-26.2 + 4.1 = -22.1^\circ\text{C}$.

4. Regular **−$0.60**

Add the 4 changes: the total change is −$2.40.

Divide by the number of days: $-\$2.40 \div 4 = -\$0.60$.

5. Rigor **4 hours**

The temperature after h hours is $9.3 - h \times 3\frac{1}{2}$.

After 3 hours: $9.3 - 3 \times 3.5 = -1.2^\circ\text{F}$, which is not below -3°F.

After 4 hours: $9.3 - 4 \times 3.5 = -4.7^\circ\text{F}$, which is below -3°F.

So the temperature is first below -3°F after 4 hours.

Re-check strip (3 items, C-NSO.2.3-R)

1. Regular -20.8°C

Total fall: $3.5 \times 5 = 17.5^\circ\text{C}$.

$-8.7 - 17.5 = -26.2$.

$-26.2 + 5.4 = -20.8^\circ\text{C}$.

2. Regular **−$0.33**

Add the 5 changes: the total change is −$1.65.

Divide by the number of days: $-\$1.65 \div 5 = -\$0.33$.

3. Regular **−$4.70**

Each share: $-\$141.00 \div 6 = -\$23.50$.

Paying $\frac{4}{5}$ of the share leaves $1 - \frac{4}{5} = \frac{1}{5}$ of it: $\frac{1}{5} \times (-\$23.50) = -\$4.70$.

Maya's balance is −$4.70.

Comparison strip (2 items, C-NSO.2.3-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **−$1.01**

Add the 5 changes: the total change is −$5.05.

Divide by the number of days: $-\$5.05 \div 5 = -\$1.01$.

2. Regular **−$5.70**

Each share: $-\$136.80 \div 4 = -\$34.20$.

Paying $\frac{5}{6}$ of the share leaves $1 - \frac{5}{6} = \frac{1}{6}$ of it: $\frac{1}{6} \times (-\$34.20) = -\$5.70$.

Ava's balance is −$5.70.

Reteach plan — C-NSO.2.3

MA.7.NSO.2.3 · Real-world problems with rational numbers

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-NSO.2.3-R

Benchmark: **MA.7.NSO.2.3** · Real-world problems with rational numbers Student code: _____ Calculator: yes

1. At 6 a.m. the temperature at a weather station was -8.7°C. The temperature then fell 3.5°C each hour for 5 hours. Later in the day it rose 5.4°C.

What was the temperature after all of these changes?

2. The price of one share of a company's stock changed by these amounts over 5 days.

−$0.25, +$1.55, −$0.70, −$2.20, −$0.05

What was the mean (average) daily change in the price?

3. A group of 6 friends owes $141.00 for a rented party tent. They split the amount owed equally, and each friend's share is recorded as a negative balance. Then Maya pays $\frac{4}{5}$ of her share.

What is Maya's balance now?

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-NSO.2.3-R

Benchmark: **MA.7.NSO.2.3** · Real-world problems with rational numbers Student code: _____ Calculator: yes

1. At 6 a.m. the temperature at a weather station was -8.7°C. The temperature then fell 3.5°C each hour for 5 hours. Later in the day it rose 5.4°C.

What was the temperature after all of these changes?

2. The price of one share of a company's stock changed by these amounts over 5 days.

−$0.25, +$1.55, −$0.70, −$2.20, −$0.05

What was the mean (average) daily change in the price?

3. A group of 6 friends owes $141.00 for a rented party tent. They split the amount owed equally, and each friend's share is recorded as a negative balance. Then Maya pays $\frac{4}{5}$ of her share.

What is Maya's balance now?

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-NSO.2.3-X

Benchmark: **MA.7.NSO.2.3** · Real-world problems with rational numbers Student code: _____ Calculator: yes

1. The price of one share of a company's stock changed by these amounts over 5 days.

+$0.85, −$2.55, −$1.55, −$0.70, −$1.10

What was the mean (average) daily change in the price?

2. A group of 4 friends owes $136.80 for a rented party tent. They split the amount owed equally, and each friend's share is recorded as a negative balance. Then Ava pays $\frac{5}{6}$ of her share.

What is Ava's balance now?

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-NSO.2.3-X

Benchmark: **MA.7.NSO.2.3** · Real-world problems with rational numbers Student code: _____ Calculator: yes

1. The price of one share of a company's stock changed by these amounts over 5 days.

+$0.85, −$2.55, −$1.55, −$0.70, −$1.10

What was the mean (average) daily change in the price?

2. A group of 4 friends owes $136.80 for a rented party tent. They split the amount owed equally, and each friend's share is recorded as a negative balance. Then Ava pays $\frac{5}{6}$ of her share.

What is Ava's balance now?

Score: ____ / 2 → **copy to your progress strip**